

ASSESSMENT

Hardware Components & System Assembly

Section – A

[Concept Application]

S [1] Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

When setting up a new office workstation, the motherboard, CPU, RAM, and storage device must be compatible with each other. I would follow these steps before purchasing the components.

1. Select the CPU: -

First, I would decide the CPU based on the office workload, such as Microsoft Office, web browsing, accounting software, video meetings, and multitasking.

I would check:

- CPU model and generation
- CPU socket type
- Power requirements

2. Select a Compatible Motherboard: -

Next, I would choose a motherboard that supports the selected CPU.

I would check:

- CPU socket compatibility
- Supported CPU generations
- Chipset compatibility
- Required expansion slots
- RAM type and maximum capacity

3. Select the RAM: -

After choosing the motherboard, I would select RAM supported by both the motherboard and CPU.

I would check:

- DDR generation, such as DDR4 or DDR5
- Supported RAM speed
- Number of RAM slots

4. Select the Storage Device: -

I would then choose an SSD or HDD according to the workstation's requirements.

I would check:

- Whether the motherboard has an M.2 slot for NVMe SSD
- Supported M.2/NVMe interface
- Available SATA ports if using a SATA SSD or HDD
- Required storage capacity

In Summary,

By checking sockets, RAM type, storage interface, and manufacturer specifications before purchasing, I can ensure all components work together properly.

S [2] Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

- **HDD (Hard Disk Drive):** Uses magnetic spinning disks. It is slower but provides large storage at a lower cost.
- **SSD (Solid State Drive):** Uses flash memory and has no moving parts. It is much faster than HDD and provides quicker boot and application loading.
- **NVMe SSD:** Uses flash memory and connects through the **PCIe interface**. It offers higher data transfer speeds and lower latency than a traditional SATA SSD.

For a workload that requires **fast application loading, quick file access, and multitasking**, I would recommend an **NVMe SSD**. Its PCIe interface provides higher bandwidth and lower latency, resulting in better overall storage performance.

For normal office work such as **Microsoft Office, web browsing, video meetings, and multitasking**, an NVMe SSD will make the system feel faster and more responsive.

S [3] Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

The GPU and cooling system are important when running graphics-intensive design software for several hours.

- **GPU:** The GPU processes graphics, images, 3D models, and visual effects. A suitable GPU provides smoother rendering and reduces the workload on the CPU.
- **Cooling System:** Continuous GPU and CPU usage produces heat. The cooling system, including **fans, heatsinks, and thermal paste**, removes this heat and keeps the components within a safe temperature range.
- **Maintaining Stability:** If the temperature becomes too high, the system may reduce performance or shut down to prevent hardware damage. Good cooling helps maintain consistent performance during long working sessions.

In Summary,

A capable GPU handles the graphics workload, while an effective cooling system prevents overheating. Together, they help maintain **stable and consistent performance** during sustained design work.

S [4] Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

Hardware compatibility is important when assembling a computer because all components must be able to communicate and work together properly.

When components come from different vendors, their **sockets, interfaces, power requirements, and supported standards** should be checked before installation.

Another possible problem is that the power supply may not provide enough power or the correct connectors for the components.

What if we ignore compatibility:

- Different CPU sockets may prevent the CPU from fitting or working.
- RAM type, such as DDR4 on a DDR5-only motherboard, will not work.
- An SSD may not work if the motherboard does not support its required interface.
- An unsuitable PSU may not provide enough power for the CPU and GPU.
- Components may not fit properly inside the case.
- An older BIOS may not support a newer CPU.

S [5] Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

The PSU should be selected by calculating the total power requirement of the CPU, GPU, motherboard, storage, and other components. I would choose a PSU with sufficient wattage and some extra capacity for future upgrades. I would also check its efficiency rating and quality.

Before powering on the PC, I would make sure the PSU is switched off and unplugged. I would check that the **24-pin motherboard connector, CPU power connector, GPU PCIe cables, and storage power cables** are properly connected and firmly secured.

I would also ensure that no loose cables are touching fans or creating a short circuit. Finally, I would inspect all connections once again and then switch on the PSU for the first boot.

Before powering on the PC, I would:

- Make sure the PSU is **switched off and unplugged**.

- Check that the **24-pin motherboard connector** is properly connected.
- Connect the **4/8-pin CPU power connector**.
- Connect the required **PCIe power cables to the GPU**.
- Confirm that the PSU voltage and connections are suitable for the components.

S [6] Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

After assembling the workstation, I would power it on and check whether it completes POST successfully.

I would confirm that the system starts normally, the display appears, and there are no unusual beep codes or error messages.

I would also check that the CPU, RAM, storage, and other hardware are detected correctly in the BIOS/UEFI.

If the system does not pass POST, I would first switch it off and check the power connections, RAM seating, CPU connection, GPU, and storage connections.

I would also check any beep codes or diagnostic LEDs to identify the faulty component.

After reseating or replacing the suspected component, I would restart the system and test POST again.

Section – B

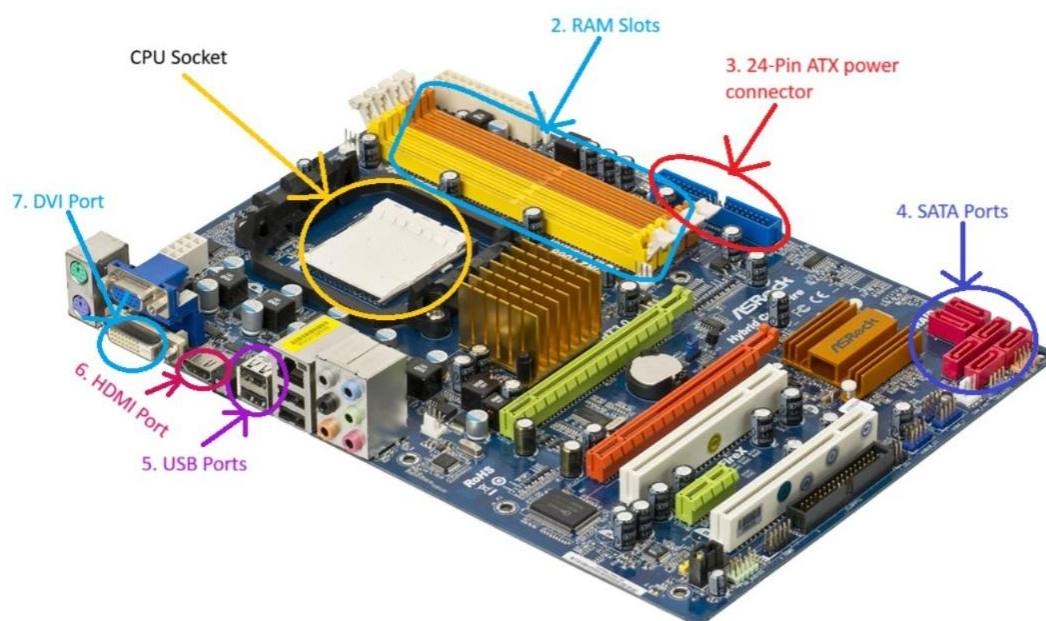
[Practical Coding Tasks]

Task 1: Component Identification & Functions

Component	Function	Make/Model or Key Specification
Motherboard	Connects all hardware components and allows them to communicate with each other.	
CPU	Processes instructions and performs calculations required by the computer.	Intel Core(TM) i5-1235U , 12 th Gen
RAM	Temporarily stores data and programs that are currently being used by the CPU.	16 GB
GPU	Processes and renders images, videos, and graphics for display.	Intel® UHD Graphics(128 MB)
Storage Device	Permanently stores the operating system, applications, and user files.	477 GB total shown in Windows
Cooling System	Removes heat from components such as the CPU and GPU to maintain safe operating temperatures.	Cooling FAN

Task 2: Motherboard Ports & Socket Identification

Port	Purpose
CPU Socket	Holds the CPU and connects it to the motherboard.
24-pin ATX Power	Supplies main power to the motherboard.
8-pin CPU Power	Supplies power to the CPU.
SATA Ports	Connect SATA HDDs, SSDs, and optical drives.
PCIe x16 Slot	Used mainly for a dedicated GPU.
RAM/DIMM Slots	Hold the system RAM modules.
USB Headers	Connect front-panel USB ports to the motherboard.
CPU Fan Header	Connects the CPU cooling fan.
M.2 Slot	Used for M.2 NVMe SSDs or other compatible devices.
PCIe x1 Slot	Used for expansion cards such as Wi-Fi or sound cards.



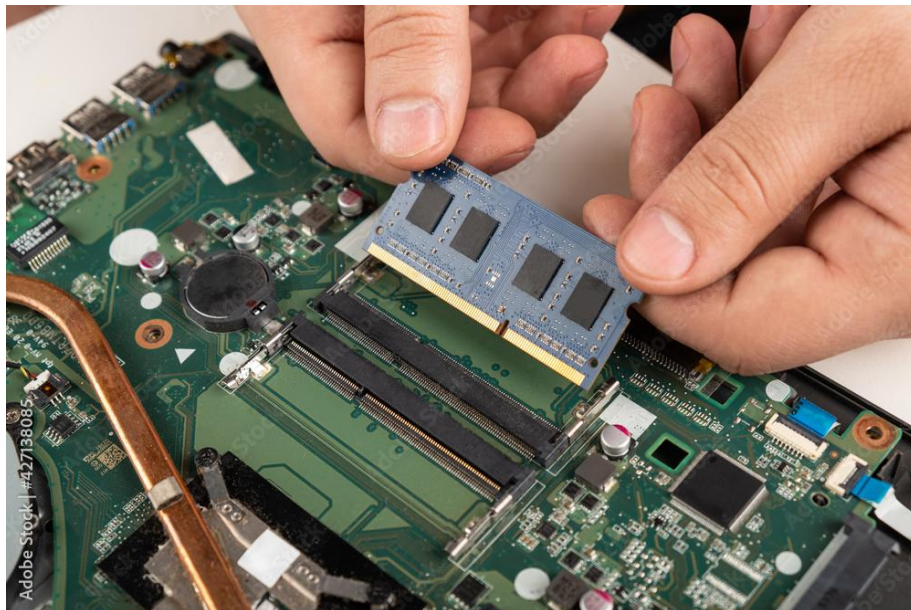
Task 3: RAM & Storage Installation

Practical Method:

This task was completed by safely removing and reinstalling the RAM module and storage device while following proper hardware handling and ESD precautions.

First of all, Install The RAM: -

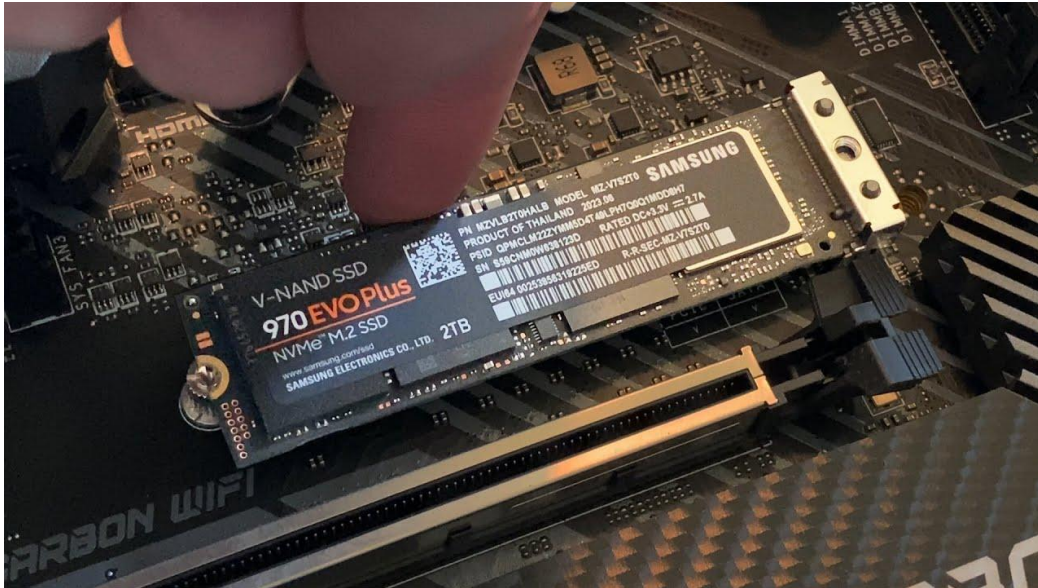
- Shut down the computer and disconnect the power.
- Locate the RAM slot on the motherboard.
- Open the side clips and carefully remove the RAM module.
- Hold the RAM by its edges and avoid touching the gold contacts.
- Align the notch on the RAM with the slot and press it down until the clips lock.
- Make sure the RAM is seated correctly and securely.



Then Storage Installation: -

M.2/NVMe SSD:

- Remove the retaining screw.
- Carefully remove the SSD from the M.2 slot.
- Insert the SSD back into the M.2 slot at an angle.
- Press it down and secure it with the retaining screw.



Safety Precautions: -

- » Disconnect power before working.
- » Use ESD protection.
- » Hold components by their edges.
- » Do not touch the gold contacts.
- » Handle the RAM and SSD carefully.

In Summary,

After installation, start the computer and check **BIOS/UEFI or Windows** to confirm that the RAM and storage device are detected correctly. This verifies that both components are properly installed and working.

Task 4: Full Desktop PC Assembly

Practical Method:

This task was completed by assembling a desktop PC using the provided motherboard, CPU, RAM, storage device, GPU, and PSU. The correct installation sequence and safety precautions were followed.

Step-by-Step Assembly

Step 1: Prepare the Work Area

Switch off and unplug the power supply. Use an anti-static wrist strap or touch a grounded metal surface before handling components.

Step 2: Install the CPU

Open the CPU socket on the motherboard. Align the CPU correctly with the socket marking and place it gently. Lock the socket mechanism.

Step 3: Install the CPU Cooler

Apply thermal paste if required and place the heatsink/cooler over the CPU. Secure it properly and connect the CPU fan cable to the CPU_FAN header.

Step 4: Install the RAM

Open the RAM slot clips. Align the notch on the RAM module with the slot and press it down until the clips lock into place.

Step 5: Install the Storage Device

For an **M.2/NVMe SSD**, insert it into the M.2 slot and secure it with the retaining screw. For a SATA drive, connect the SATA data and power cables.

Step 6: Install the Motherboard

Place the motherboard inside the case and align it with the standoffs and rear I/O area. Secure it using the appropriate screws.

Step 7: Install the GPU

Insert the GPU into the **PCIe x16 slot** and secure it to the case. Connect the required PCIe power cable if needed.

Step 8: Install the PSU

Place the PSU in the case and secure it with screws. Connect the **24-pin motherboard power** and **4/8-pin CPU power** connectors.

Step 9: Connect Remaining Cables

Connect storage power/data cables, GPU power cables, CPU fan, and front-panel connectors. Check that all cables are firmly connected.

Step 10: Final Safety Check

Check that all components are properly seated and that no loose screws or cables are inside the case. Make sure cables are away from the fans.

POST Verification

Step 11: Power On the PC

Connect the monitor, keyboard, and power cable. Switch on the PSU and start the PC.

Step 12: Check POST

Confirm that the system displays the BIOS/UEFI screen and detects the **CPU, RAM, storage, and GPU**. Check for any beep codes, warning messages, or diagnostic LED errors.

In Summary,

The assembly is successful when all components are correctly installed and the computer powers on and completes POST (Power-On Self-Test) without errors.

Section – C

[Mini Capstone Project]

Task = Workstation Deployment & Boot-Failure Diagnosis

Objective :-

This project focuses on assembling or inspecting a workstation, identifying possible boot failures, checking hardware compatibility, and verifying that all components are installed correctly.

1. Three Hardware Components to Check First

If a newly assembled PC fails to boot, I would check these three components first:

1. PSU (Power Supply Unit):

I would check whether the PSU is switched on and whether the motherboard and CPU power connectors are properly connected. Without proper power, the system will not start.

2. RAM:

I would check whether the RAM is correctly seated in the DIMM slot. Loose or incorrectly installed RAM can cause POST failure, no display, or beep/LED errors.

3. CPU and Motherboard:

I would check that the CPU is correctly installed and that the motherboard supports that CPU. A CPU installation or compatibility problem can prevent the system from completing POST.

2. Motherboard–CPU Compatibility

Motherboard and CPU compatibility is important because the CPU must match the motherboard's **socket and supported CPU generation**.

For example, if the inspected motherboard has an **AM5 socket**, the CPU must be an AM5-compatible processor.

I would also check the motherboard manufacturer's CPU support list and BIOS requirements before installation. An incompatible CPU may result in the system failing to boot or complete POST.

3. RAM and Storage Verification

After installing the RAM, I would make sure the module is fully seated and the locking clips are secured.

I would then enter **BIOS/UEFI** and confirm that the correct amount of RAM is detected.

For storage, I would check that an **NVMe SSD is properly installed in the M.2 slot** or that a SATA drive has both its data and power connections attached.

BIOS/UEFI should detect the storage device, and after booting into Windows, it should also appear in **Disk Management/File Explorer**.

4. Hardware-Installation Checklist

Before powering on each workstation, I would follow this checklist:

- » Check CPU installation, CPU socket compatibility, and CPU cooler connection.
- » Confirm RAM is correctly seated and locking clips are secured.
- » Confirm SSD/HDD is properly connected and detected.
- » Check the 24-pin motherboard and 4/8-pin CPU power connectors.
- » Check GPU installation and required power cables.
- » Check all components for loose connections before the first boot.
- » Power on and confirm successful POST and BIOS/UEFI hardware detection.

5. Why Use SSD Instead of HDD?

For the computer lab, I would use **SSDs instead of HDDs** because modern office workstations need quick boot times, faster application loading, and responsive multitasking.

SSDs have no moving mechanical parts, so they are generally quieter and more resistant to physical shock than HDDs. An **NVMe SSD** can provide even higher performance through the PCIe interface, making it suitable for a lab where many users may regularly run office applications, browse the web, access files, and perform multitasking.

Section – D

[AI-Augmented Learning]

STEP 1 – Build with AI(ChatGPT)

AI Prompt Used: -

“Act as a junior IT support guide. Create a simple troubleshooting checklist for a newly assembled desktop computer that does not boot. Start with the most basic and important checks, then move to RAM, motherboard, CPU compatibility, storage, and PSU. Include what the technician should check at each step and how BIOS/POST can help identify the problem. Also explain how this checklist can be used to troubleshoot the boot and hardware detection problems reported at ABC Solutions.”

Original AI Checklist :-

1. Check the power supply and power connections.
2. Check the motherboard for indicator lights or error signals.
3. Reseat and test the RAM module.
4. Verify CPU and motherboard compatibility.
5. Check the storage device and its connections.
6. Check BIOS/UEFI to confirm hardware detection.
7. Start the system again and verify successful POST.

STEP 2 – TEST & DEBUG (WITHOUT AI)

The AI checklist was useful, but I reviewed the steps and changed their order to make the troubleshooting process safer and more practical.

Corrected Checklist :-

1. Check the PSU and main power connection.
2. Check the 24-pin motherboard and CPU power cables.
3. Check whether the CPU is correctly installed and compatible with the motherboard.

4. Reseat the RAM and check whether it is detected.
5. Check the GPU connection if a dedicated GPU is installed.
6. Check the storage device and its connections.
7. Start the PC and check POST/BIOS for error messages.

What I Changed and Why: -

I moved the basic power and motherboard connection checks to the beginning because the PC cannot start without proper power. I placed CPU, RAM, and GPU checks before storage because these components can prevent the system from completing POST. I also kept the BIOS/POST check at the end to confirm whether the hardware problem has been resolved.